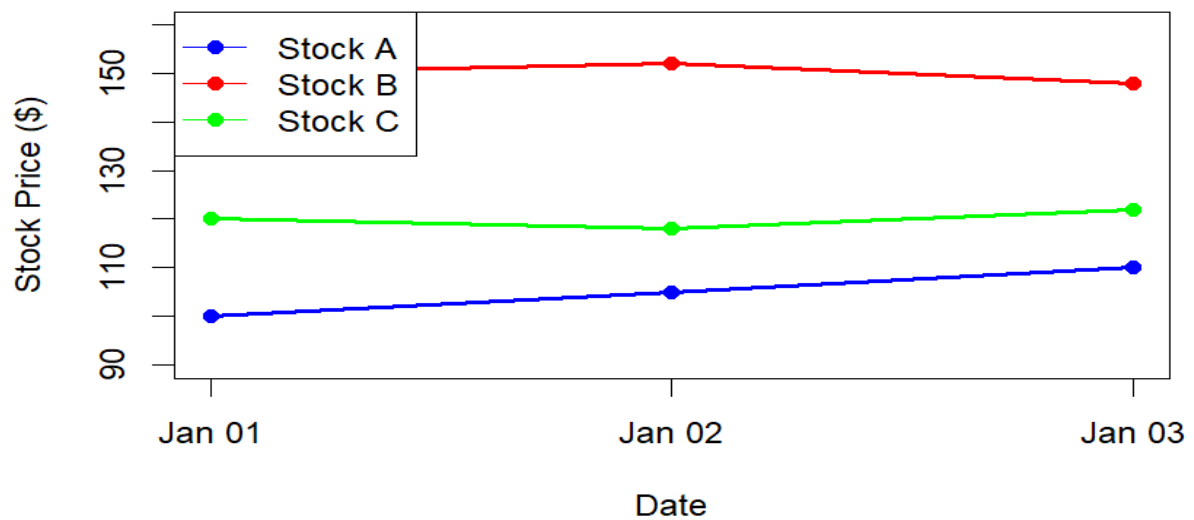


LAB EXPERIMENTS

SET-16

QUESTION-1

Stock Price Analysis



CODE:

```
# Stock Price Data
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))

stock_A <- c(100, 105, 110)
stock_B <- c(150, 152, 148)
stock_C <- c(120, 118, 122)

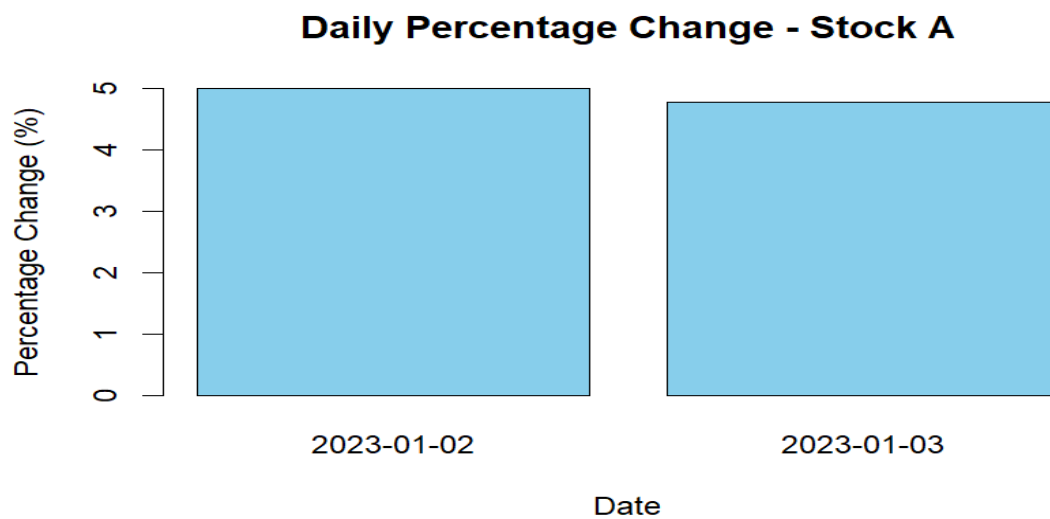
# Line Chart
plot(date, stock_A,
      type = "o",
      col = "blue",
      pch = 16,
      lwd = 2,
```

```

xlab = "Date",
ylab = "Stock Price ($)",
main = "Stock Price Analysis",
ylim = c(90, 160))
lines(date, stock_B,
      type = "o",
      col = "red",
      pch = 16,
      lwd = 2)
lines(date, stock_C,
      type = "o",
      col = "green",
      pch = 16,
      lwd = 2)
legend("topleft",
      legend = c("Stock A", "Stock B", "Stock C"),
      col = c("blue", "red", "green"),
      lty = 1,
      pch = 16)

```

QUESTION-2



CODE:

```
# Stock A Prices

stock_A <- c(100, 105, 110)

# Calculate Daily Percentage Change

percentage_change <- c(
  NA,
  ((stock_A[2] - stock_A[1]) / stock_A[1]) * 100,
  ((stock_A[3] - stock_A[2]) / stock_A[2]) * 100
)

print(percentage_change)

# Remove first NA value

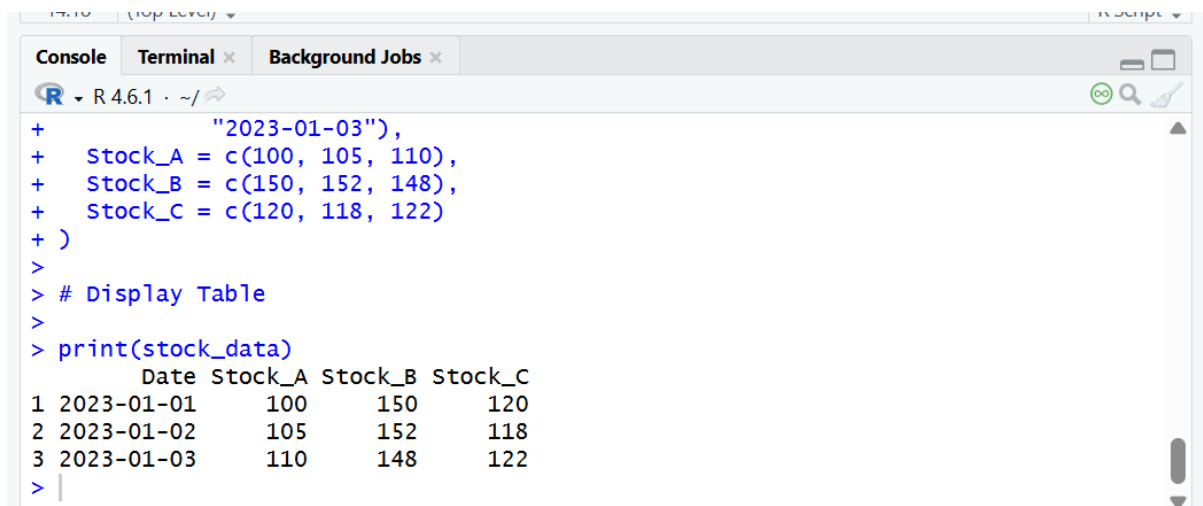
change <- percentage_change[2:3]

days <- c("2023-01-02", "2023-01-03")

# Bar Chart

barplot(change,
  names.arg = days,
  col = "skyblue",
  xlab = "Date",
  ylab = "Percentage Change (%)",
  main = "Daily Percentage Change - Stock A")
```

QUESTION-3



The screenshot shows an R console window with the following code and output:

```
R 4.6.1 · ~/ · ↗
+ "2023-01-03"),
+   Stock_A = c(100, 105, 110),
+   Stock_B = c(150, 152, 148),
+   Stock_C = c(120, 118, 122)
+ )
>
> # Display Table
>
> print(stock_data)
      Date Stock_A Stock_B Stock_C
1 2023-01-01     100     150     120
2 2023-01-02     105     152     118
3 2023-01-03     110     148     122
> |
```

CODE:

```
# Stock Price Dataset

stock_data <- data.frame(

  Date = c("2023-01-01",
           "2023-01-02",
           "2023-01-03"),

  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

# Display Table

print(stock_data)
```